

What makes a TEDTalk go viral?

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Problem Statement

Can we predict which **TED Talk**
can go viral based
on its intrinsic characteristics?

Why it matters?

- Maximize the reach of ideas
- Optimize limited resources
- Guide content decisions

The Data

TED

6,396 talks

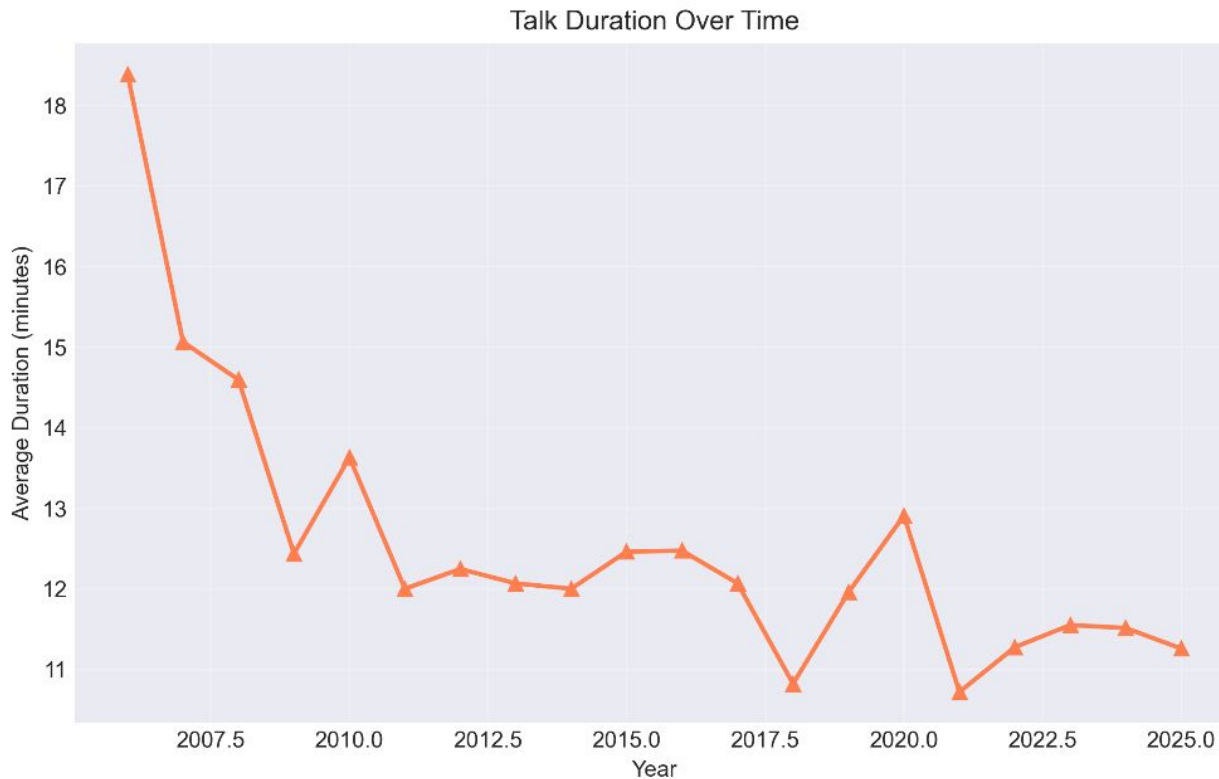
WIKIPEDIA
The Free Encyclopedia

5,674 speakers

TED Talks are getting shorter

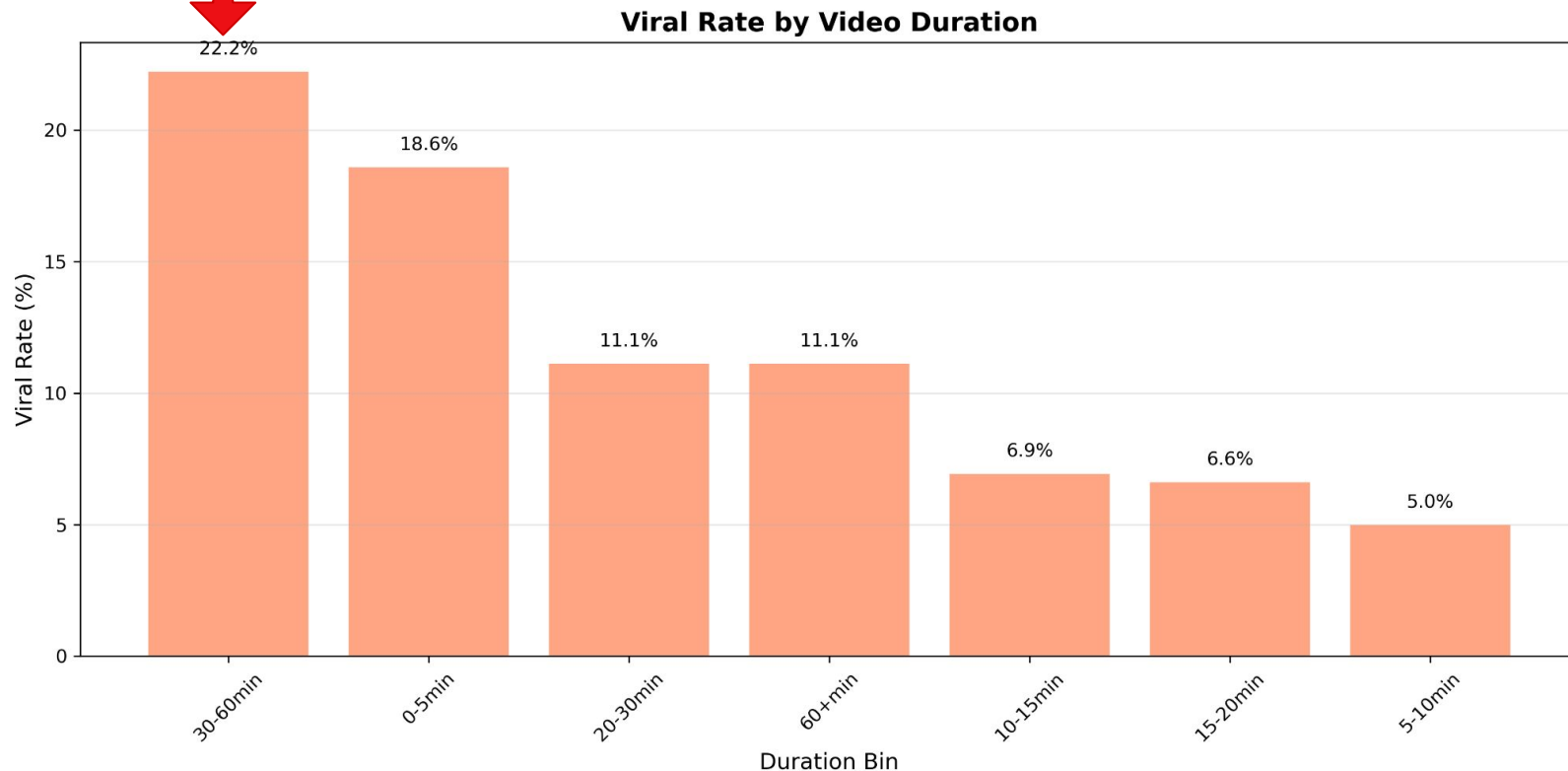
Platform strategy is shifting toward brevity...

but does audience demand follow?

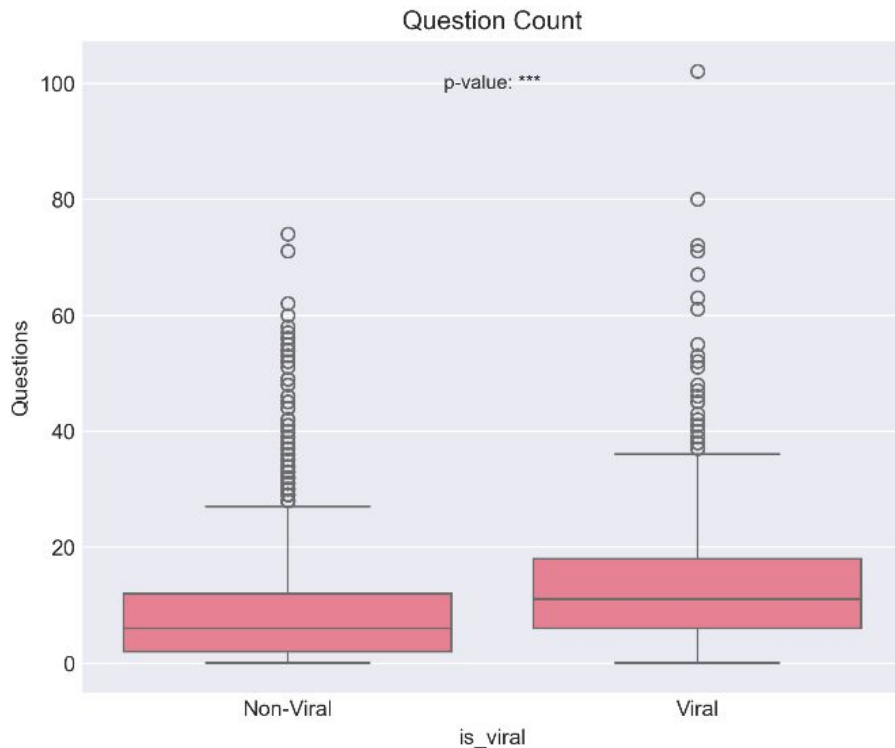


Depth > Brevity

Duration Distribution



Viral Speakers Ask More Questions

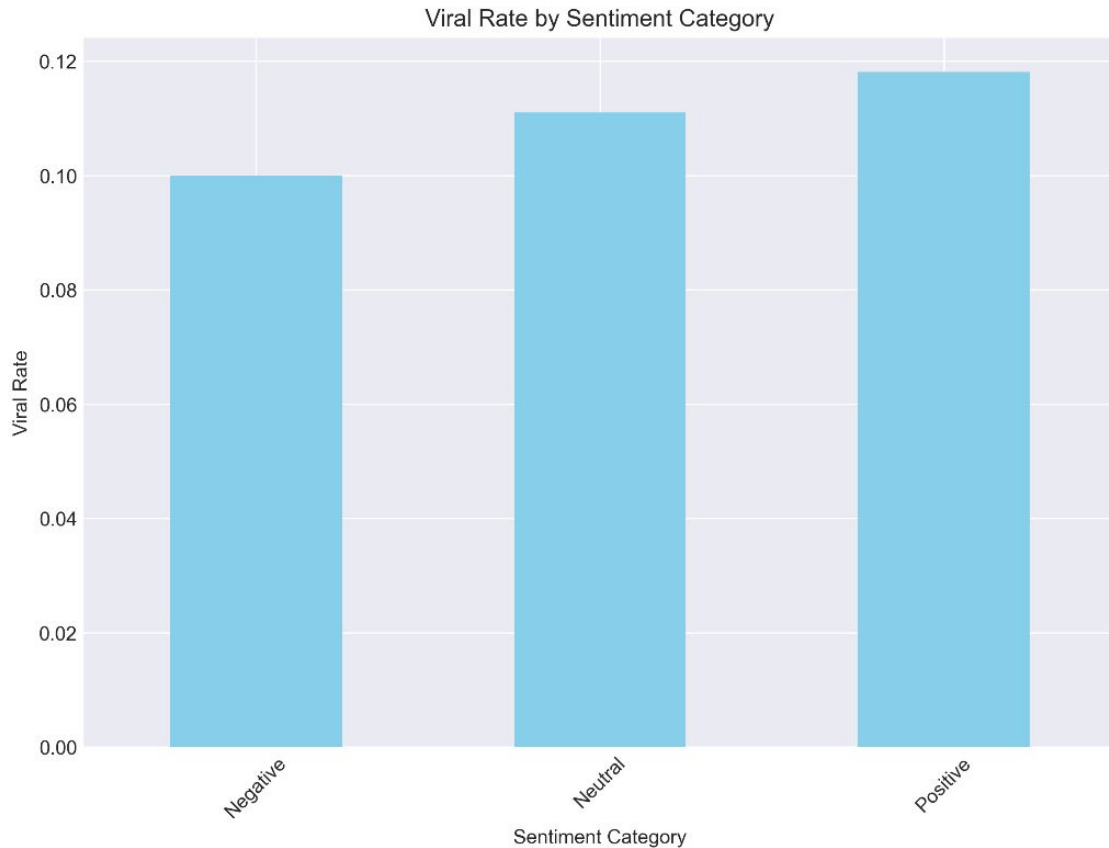


Significantly higher
use of rhetorical
questions.

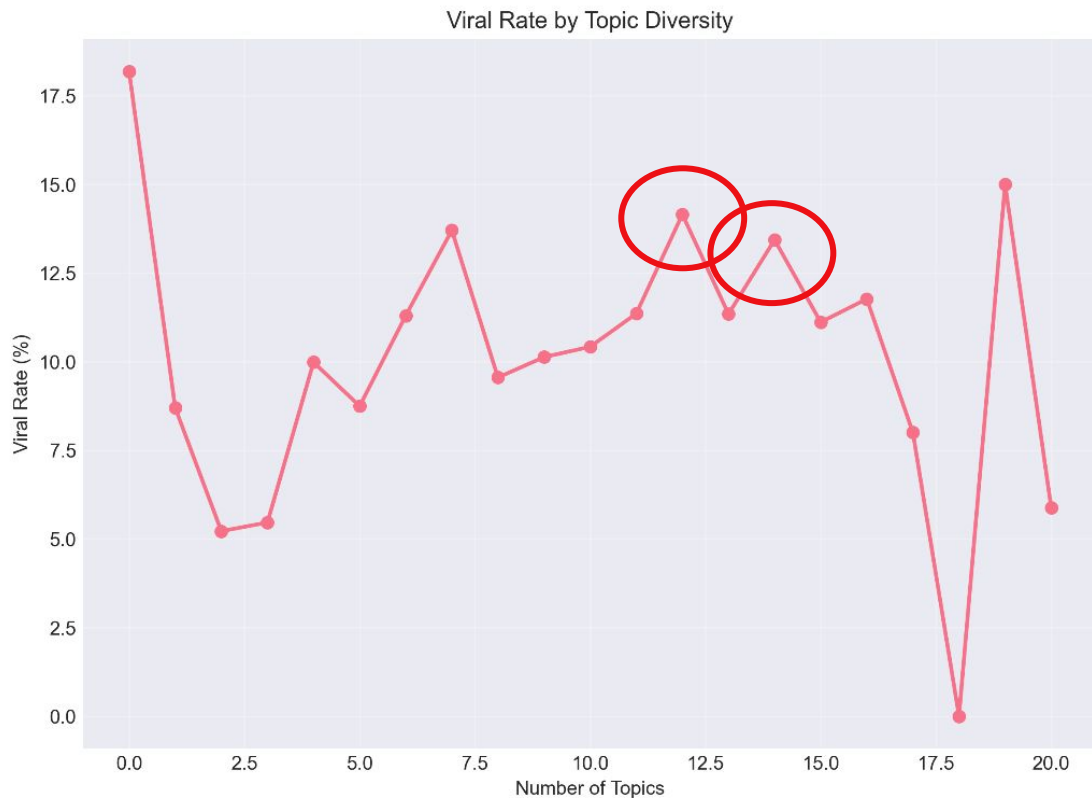
Successful
speakers engage
rather than lecture.

The Positivity Bias

Positive Talks have a higher probability of going viral...



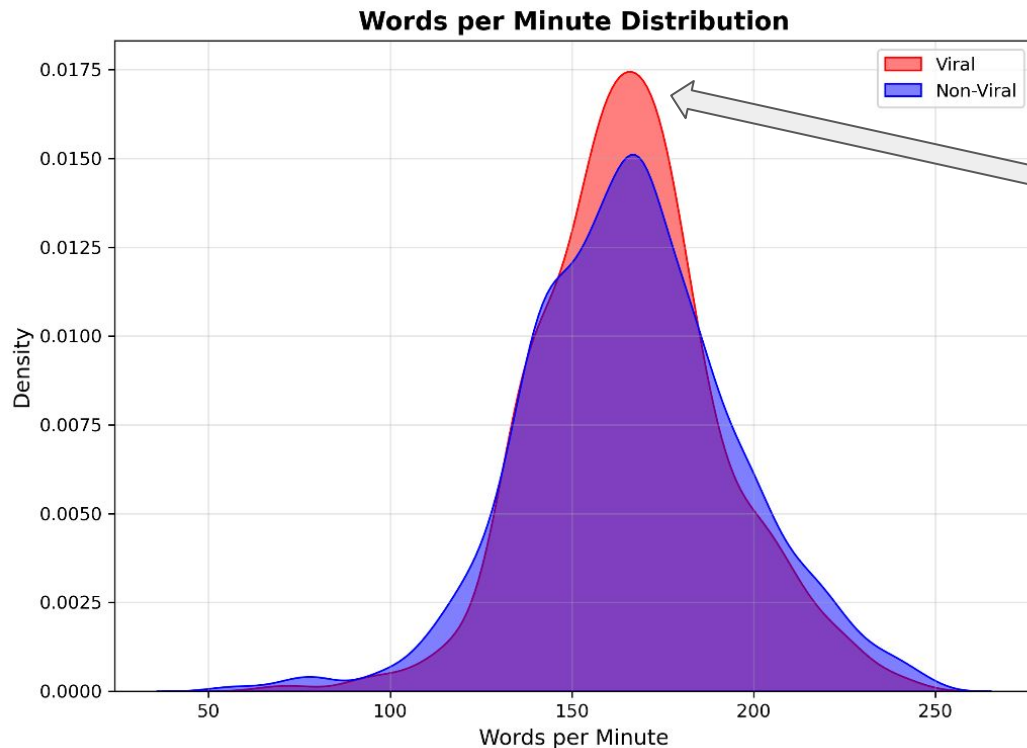
Topic Diversity



Talks bridging 10-15 topics maximize **reach**.

Viral Speakers Maintain High Energy

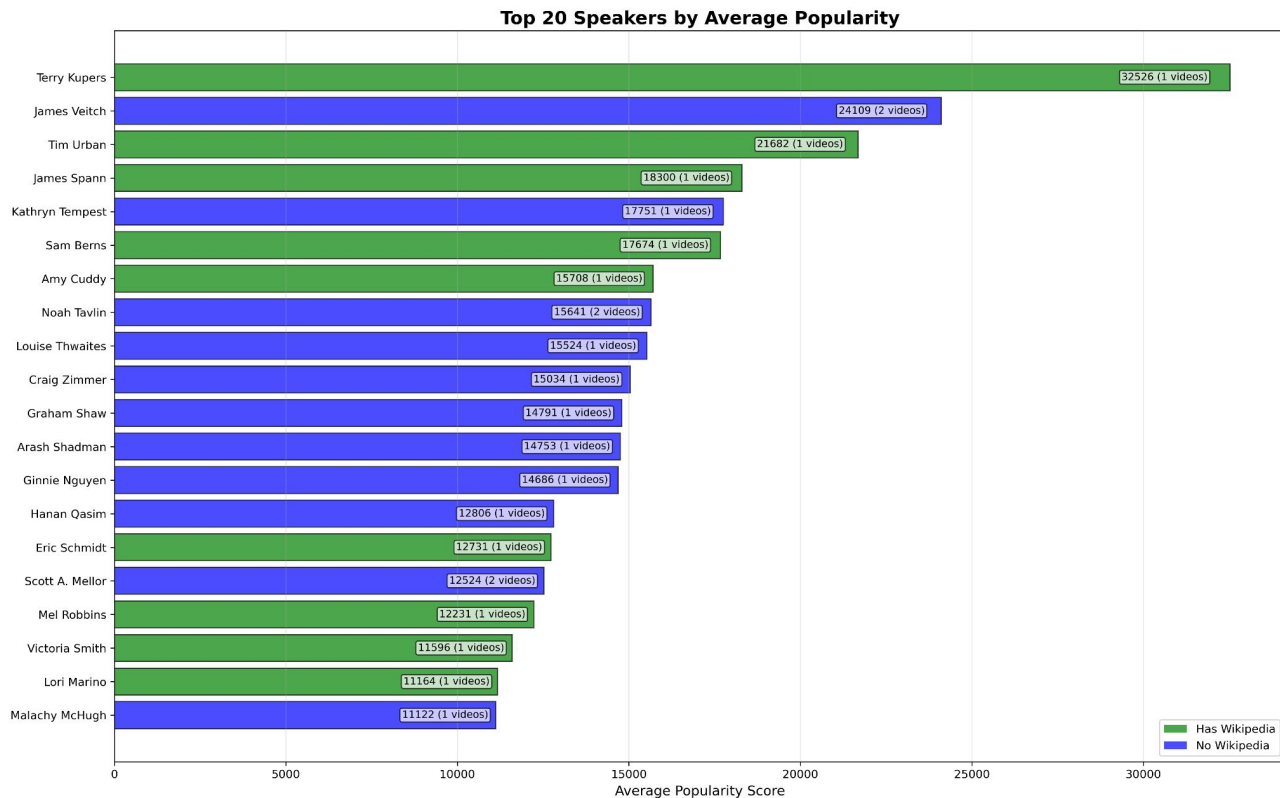
Consistency
matters.



Target Pace:
160-175
WPM
(Dynamic &
Fast)

Speaker Feature Analysis

Who is the most popular?



Average Popularity score for speakers with

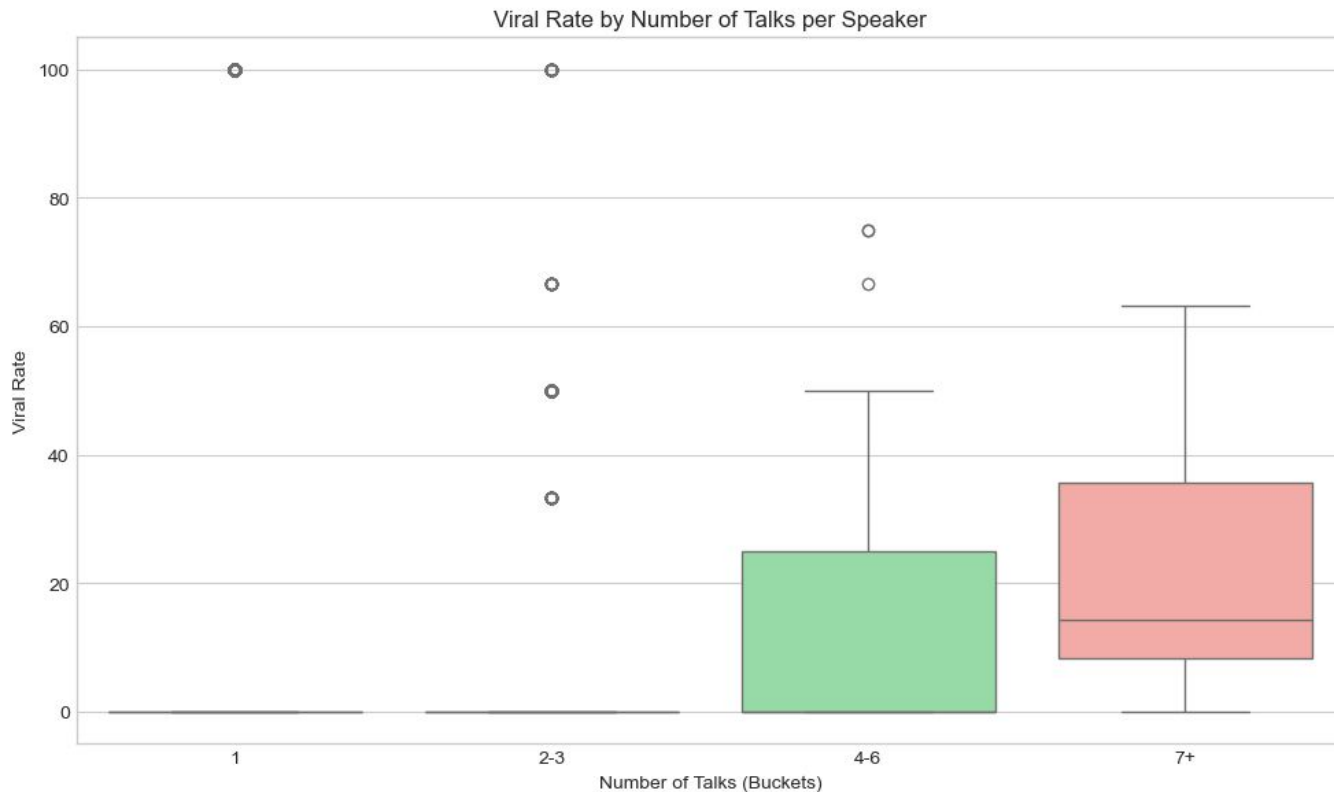
No Wikipedia

1504.6

Wikipedia

1262.9

TED experience = higher viral rates



EDA Conclusion

- No discriminative features for virality
- Distribution overlaps
- Viral talks tend to be moderately long, engaging
- Speakers with **TED** experience have higher viral rates

Predictive Modeling

Finding the right Model

Problem: Viral talks are rare (only 10%)

Solution: SMOTE to generate synthetic viral examples, teaching the model what 'success' looks like.

Model Performance Comparison

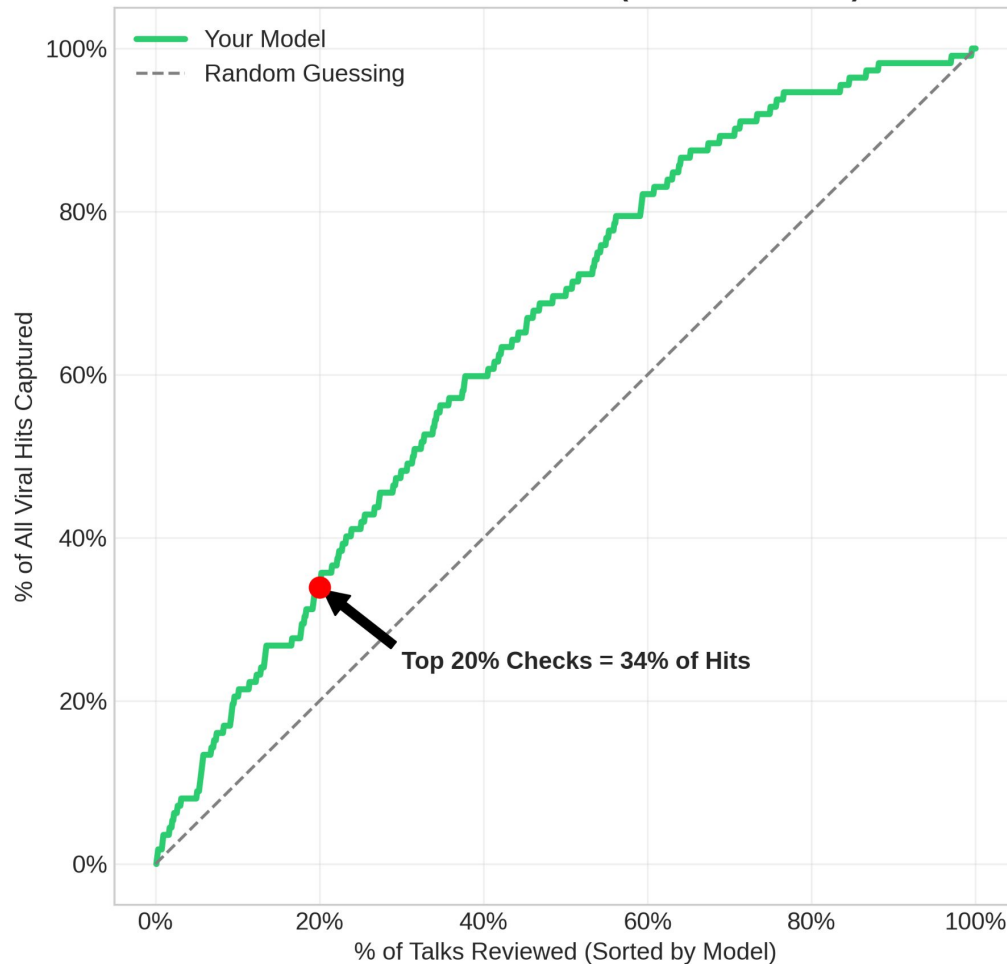
Metric	Random Baseline	Logistic Reg.	XGBoost (Final)	Lift
PR-AUC	0.10	0.18	0.21	2.1x
Recall	10%	52%	73%	7.3x
Precision	10%	15%	18%	1.8x
F1-Score	0.10	0.23	0.29	2.9x

XGBoost outperforms all.

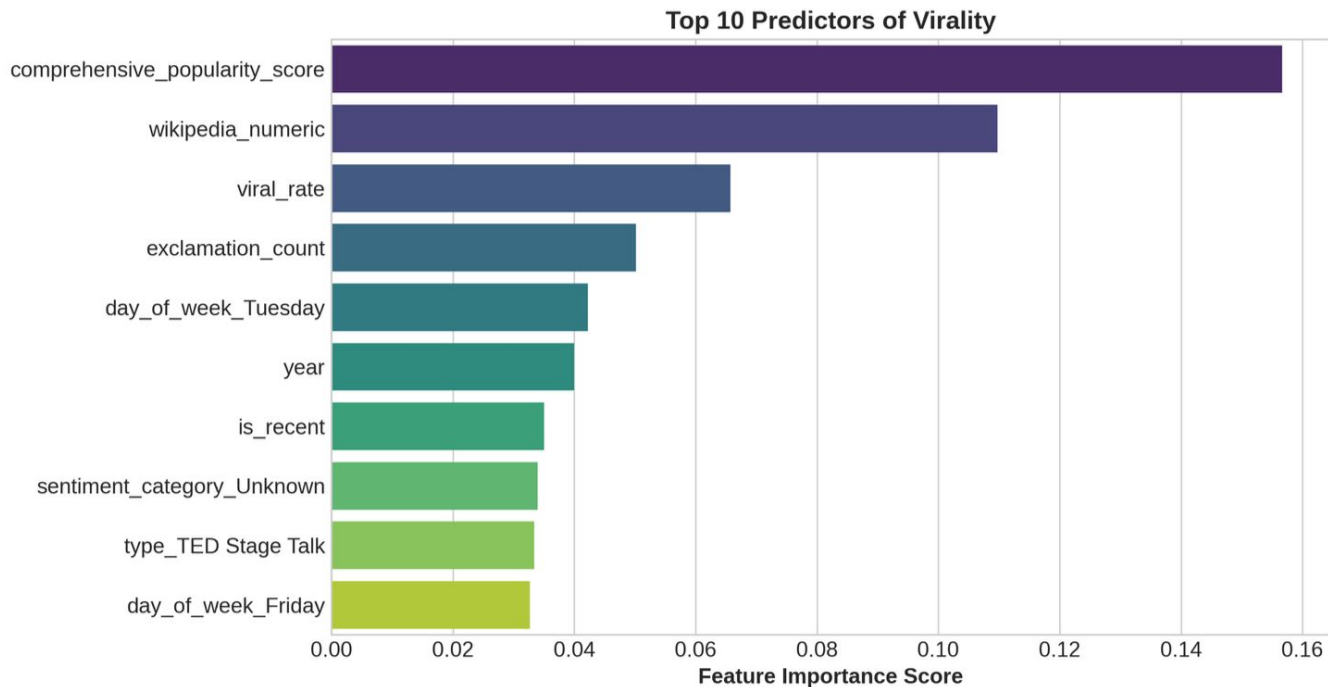
Higher **Recall** is our priority.

Efficiency: We Find Hits Faster

3x Faster Discovery



The 3 Drivers of Virality



Platform History

Emotional Intensity

Strategic Timing

Conclusion

Go Deeper. Be engaging.

